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# hCRI Companion – User Manual

Covers app version 1.2 · Last updated October 3, 2026 · [Source on GitHub ↗](#)

## 01 Overview

hCRI Companion is an independent Android app that connects to a Hopoocolor HPCS-series spectrometer over Bluetooth Low Energy, reads the full spectrum off the sensor, computes the same colorimetric values hCRI.io itself computes, and uploads a reading to your hCRI.io account in one tap – no cable, no CSV export, no re-typing numbers off the meter's own tiny screen.

Every CCT, Duv, Ra, Rf, Rg, and R1–R15 value shown in the app is computed with the exact same algorithm hCRI.io's report page uses, so a reading never disagrees with itself between your phone and your account.

### Supported meters

HPCS-310

HPCS-310P

HPCS-330

HPCS-330P

HPCS330Pro

The app detects which model you're connected to from its advertised Bluetooth name and reads its data with that model's own field layout automatically – nothing to configure. The full, up-to-date list (including which models are independently verified against a real captured reading) is always on the app's own Settings → About section.

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## 02 Requirements

- Android 8.0 (Oreo) or newer
- Bluetooth Low Energy (BLE) support
- A paired Hopoocolor HPCS-series meter, powered on and in range
- An hCRI.io account, if you want to upload readings (not required just to take and view them)

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## 03 Installing the app

hCRI Companion isn't on the Play Store – it's a direct install, the same way most spectrometer and BLE utilities in this space are distributed.

### 1 Download the APK

From the [Companion download page](#), tap Download APK. Android downloads the file like any other.

### 2 Allow installs from this source

Opening the downloaded file for the first time, Android will ask to confirm installing from an unknown source – this is expected for any app distributed outside the Play Store. Allow it for your browser or Files app.

### 3 Install and open

Finish the install prompt, then open hCRI Companion from your app drawer.

#### Updating later

There's no in-app auto-update. Re-download and install the APK the same way to get a newer build – it installs over the existing app and keeps your saved readings, account, and settings.

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## 04 Connecting your hCRI.io account

You only need to do this once. From the `Settings` tab:

1 **Enter your hCRI.io username**

The same username you sign in with at hcri.io.

2 **Enter your hCRI.io API token**

Find or generate this from your hCRI.io account. The field is masked while typing.

3 **Tap Save**

The token is stored in your device's secure keychain (Android Keystore) – not in plain text, and not readable by any other app.

Once saved, the fields switch to a locked, read-only view with a `Forget Credentials` button – a deliberate extra step before replacing a saved key, so a stray keystroke can't silently overwrite a working token.

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## 05 Pairing with your meter

On the `Main` tab, tap `Connect to Meter`. The app scans for a few seconds and connects automatically if exactly one supported meter is in range.

If more than one matching meter is found, a picker appears so you choose which one – look for its advertised name and signal strength. A small `≡ Switch meter` icon next to the status row lets you reopen that same list later without a fresh scan.

### Meter won't connect?

Tap `Meter won't connect? Reset connection`, underneath the Connect button. This clears a stale Bluetooth connection left over from a previous session – the most common reason a meter seems stuck, and something that used to require power-cycling the meter itself.

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## 06 Taking a reading

Once connected, tap `Take Reading`. The meter takes its exposure, and a short haptic buzz confirms the moment it lands (a different pattern if it fails) – handy when you're not looking directly at the screen. The spectrum, CCT, Duv, Ra, and the rest of your chosen measurements appear immediately after.

Every completed reading is saved automatically to [History](#) – there's nothing extra to do to keep it.

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## 07 The Main tab

Main is where you connect, take readings, and review the latest one. Its result card always shows the same layout – a grid of measurements, the three swipeable charts, and the action button directly underneath them – whether or not a reading has happened yet (before your first reading, everything just reads zero/blank, so nothing jumps around once real numbers appear).

### Measurement grid

Shows CCT, Ra (CRI), Duv, Lux, Rf, R9, and Rg by default. Which measurements appear here, and in what order, is fully customizable from [Settings](#) – including any individual R1–R15 sample and the raw CIE x/y coordinates.

### Spectrum / Chrom / R-Values

Swipe (or tap the dots) between three views of the same reading:

- **Spectrum** – the full wavelength-by-wavelength power distribution, colored by wavelength, plus a collapsible list of every raw nm/value pair.
- **Chrom** – the CIE 1931 chromaticity diagram, with your reading's x/y point plotted against the real spectral locus and the Planckian (blackbody) curve.
- **R-Values** – the full CRI R1–R15 breakdown as a color-coded bar chart, the same chart hCRI.io's own report page uses.

### Uploading

Tap `Upload to hCRI.io` to send the current reading. A small green checkmark appears next to the button once it succeeds – there's no popup to dismiss on a routine upload. If an upload fails, an alert explains what happened; check the [Logs tab](#) for the server's own response if you need more detail.

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## 08 The Data tab

A closer look at the current reading, plus the controls for naming and sending it:

- **Upload Title** – type a custom title, or leave it blank to use the auto-generated username + timestamp + device label.
- **Upload Preview** – the exact wavelength/value payload about to be sent, so you can eyeball the range and point count before it goes anywhere.
- **R1–R15, Chromaticity, and Device Info** – full numeric detail, including the meter's own firmware version and integration time.
- **Upload to hCRI.io** and **Share CSV** – the same upload button as Main (with the same inline checkmark), plus exporting this one reading as a CSV file through Android's share sheet.

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## 09 The History tab

Every reading you've ever taken, grouped by day, newest first. From here you can:

- **View** a past reading's full measurement grid, spectrum, chromaticity, and R-values exactly as they looked when it was taken.
- **Rename** a reading – renaming before upload sends it to hCRI.io under that new title.
- **Re-upload** a reading at any time, including ones already sent.
- **Select** multiple readings and upload them to hCRI.io in one batch, instead of opening and sending each individually.
- **Share** a single reading, or your entire history, as CSV.
- **Delete** a reading you no longer want – this only removes it from your phone; a copy already uploaded to hCRI.io stays there until you delete it from your account separately.

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## 10 The Logs tab

A running debug log of connect/disconnect/measure/upload activity, each line timestamped in your phone's own local time. Useful for troubleshooting, or to attach to a bug report.

- **Share Debug Log** – sends the current log through Android's share sheet (email, messaging, wherever you'd like to send it).
- **Clear** – empties the log.

Turning on **Verbose logging** in Settings adds the low-level Bluetooth hex traffic and the full raw result body to this log – off by default, since the standard log already covers ordinary connection and measurement status.

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## 11 Settings

### hCRI.io account

Your username and API token – see [Connecting your hCRI.io account](#) above.

### Appearance

Light, Dark, or System (the default) – follows your phone's own light/dark setting unless overridden here.

### Stay connected in background

**On by default.** Keeps the meter connected when you switch away from the app or lock the screen, so coming back doesn't need a fresh reconnect. Turn this off to disconnect the meter whenever the app isn't in the foreground instead – saves the meter's own battery, at the cost of reconnecting (a few seconds) every time you come back. See [Background & reconnect behavior](#) for the full picture.

### Keep screen awake while connected

Prevents the screen from locking during a session with the meter connected. Uses more battery – off by default.

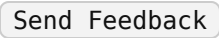
### Verbose logging

Adds the full wavelength-by-wavelength and Bluetooth hex detail to the debug log – see [The Logs tab](#).

### Main Screen Measurements

Check which measurements show on the Main tab's result card, and hold the  handle to drag a row and reorder it.  restores CCT, Ra, Duv, Lux, Rf, R9, Rg in that order.

### About

The current list of supported meter models (and which are independently verified), a  button that opens a pre-addressed email, and a link back to this Companion page.

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## 12 Background & reconnect behavior

By default, switching away from hCRI Companion – the app switcher, locking your screen, a phone call – leaves the meter connected (`Stay connected in background`, on by default in Settings). Coming back to the app picks up right where you left off, with no reconnect delay.

If you'd rather the app disconnect the meter whenever it isn't the one on screen (trading a short reconnect delay for a little extra meter battery life), turn that setting off. Either way, coming back to the foreground with nothing connected always tries to reconnect automatically – you shouldn't need to tap Connect yourself just because the meter dropped on its own while you were away.

Sharing a CSV or the debug log briefly hands control to Android's own share sheet, which looks like backgrounding the app – this is recognized and never triggers a disconnect either way, since you're still right there using the app.

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## 13 Troubleshooting

### Meter won't connect, or seems stuck

Tap `Meter won't connect? Reset connection` on the Main tab (shown while disconnected). This clears a leftover Bluetooth connection from a previous session – by far the most common cause, and something that used to require power-cycling the meter.

### App crashed, or won't open

A crash lands on a "Try Again" screen rather than kicking you back to your home screen. If the app won't stay open at all, force-close it from Android's recent-apps view and reopen it – this app recovers from a mid-session crash cleanly.

### A reading looks wrong

Double-check the correct meter model connected (shown next to the status dot on Main) – each model has its own field layout, and the wrong one can occasionally still connect if its advertised name happens to be ambiguous. If a value still looks off, check Settings → About for whether that model is marked independently verified.

### Still stuck?

From the Logs tab, tap `Share Debug Log` and send it along with a description of what happened to [fizzgig@hcri.io](mailto:fizzgig@hcri.io), or use `Send Feedback` on the About section of Settings.

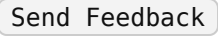
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## 14 Privacy

hCRI Companion doesn't collect, store, or transmit anything about you on its own – no analytics, no tracking, no third-party server in the loop. A reading lives only on your device until you tap Upload, which sends it straight to your own hCRI.io account over an encrypted connection and nowhere else. Full details are in the [Privacy Policy](#) on the Companion page.

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## 15 Feedback & support

Questions, bug reports, or feature requests are always welcome at [fizzgig@hcri.io](mailto:fizzgig@hcri.io) – the same address the app's own  button (Settings → About) opens for you.